American International University- Bangladesh (AIUB)
Faculty of Engineering (EEE)

Course Name :	Microprocessor and Embeded System	**Course Code :**	
Semester :		**Sec : N**	
Lab Instructor :	Md Sajid Hossain		

Experiment No :	01
Experiment Name :	Familiarization with a microcontroller, the study of blink test and implementation of a traffic control system using microcontrollers.

Submitted by (NAME):	**Kotha Dhar**	**Student ID: 23-53561-3**	

Group Members		**ID**	**Name**
	1.	23-50721-1	Kazi Al Irfan
	2.	23-50753-1	Md. Sabbir Hossain Rohan
	3.	23-53499-3	Md. Najir Ahmed
	4.	23-50567-1	Md. Raihanul Islam
	5.		
	6.		
	7.		

Performance Date :	**18.02.2026**	**Due Date :25.02.2026**	

Marking Rubrics (to be filled by Lab Instructor)

Category	Proficient [6]	Good [4]	Acceptable [2]	Unacceptable [1]	Secur Mark
Theoretical Background, Methods & procedures sections	All information, measures and variables are provided and explained.	All Information provided that is sufficient, but more explanation is needed.	Most information correct, but some information may be missing or inaccurate.	Much information missing and/or inaccurate.	
Results	All of the criteria are met; results are described clearly and accurately;	Most criteria are met, but there may be some lack of clarity and/or incorrect information.	Experimental results don't match exactly with the theoretical values and/or analysis is unclear.	Experimental results are missing or incorrect;	
Discussion	Demonstrates thorough and sophisticated understanding. Conclusions drawn are appropriate for analyses;	Hypotheses are clearly stated, but some concluding statements not supported by data or data not well integrated.	Some hypotheses missing or misstated; conclusions not supported by data.	Conclusions don't match hypotheses, not supported by data; no integration of data from different sources.	
General formatting	Title page, placement of figures and figure captions, and other formatting issues all correct.	Minor errors in formatting.	Major errors and/or missing information.	Not proper style in text.	

Writing & organization	Writing is strong and easy to understand; ideas are fully elaborated and connected; effective transitions between sentences; no typographic, spelling, or grammatical errors.	Writing is clear and easy to understand; ideas are connected; effective transitions between sentences; minor typographic, spelling, or grammatical errors.	Most of the required criteria are met, but some lack of clarity, typographic, spelling, or grammatical errors are present.	Very unclear, many errors.	
Comments:				Total Marks (Out of):	

Title: Familiarization with a microcontroller, the study of blink test and implementation of a traffic control system using microcontrollers.

Abstract:

The purpose of this experiment is to introduce the basic concepts of microcontrollers through hands-on practice using the Arduino platform. The experiment focuses on understanding microcontroller operation, performing an LED blink test, and developing a simple traffic light control system. This work is significant as it provides foundational knowledge of embedded systems and demonstrates the interaction between software programming and hardware control. The methodology includes writing and uploading programs using the Arduino IDE and interfacing LEDs with resistors to construct the required circuits. Initial testing is carried out through a basic LED blinking program to ensure correct setup, followed by the implementation of a traffic control sequence using red, yellow, and green LEDs with specific timing delays. Key observations include accurate digital output control and proper execution of timing sequences. The scope of the experiment is limited to delay-based control techniques and does not involve interruptions or hardware timers. The successful operation of both the blink test and traffic control system confirms the effectiveness of the approach and enhances practical understanding of microcontroller-based embedded applications.

Objective:

The objective of this experiment is to get familiarized with Microcontrollers. Besides, the specific

objectives of this experiment are to-

• Make an LED blink using an Arduino and its delay function.

• Implement a traffic control system using Arduino and its delay function and LEDs.

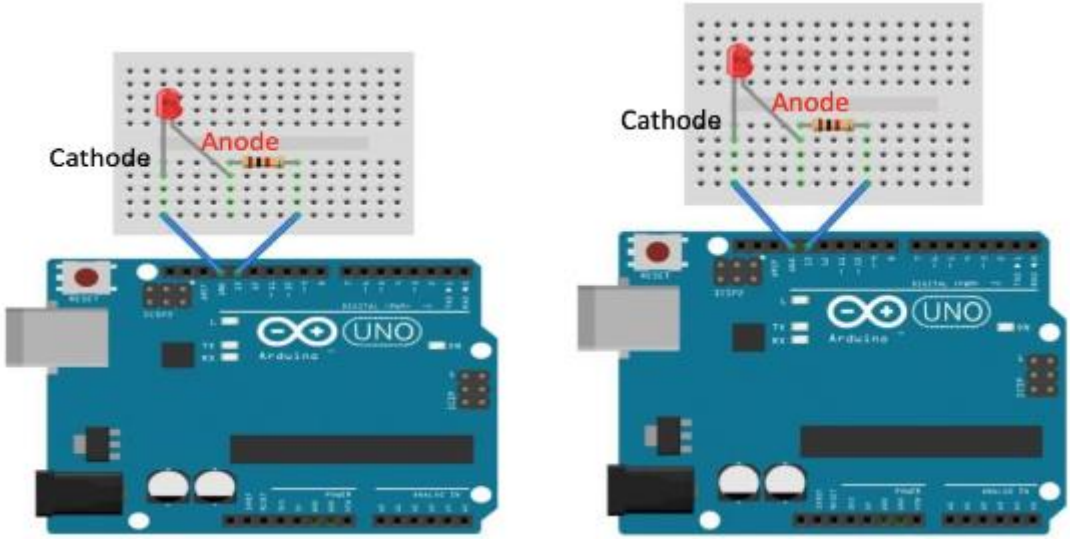
Theory and Methodology:

Arduino is an open-source platform used for creating interactive electronics projects. Arduino consists of both a programmable microcontroller and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the microcontroller board. Arduino Uno also doesn't need a hardware circuit (programmer/ burner) to load a new code into the board. We can easily load a code into the board just using a USB cable and the Arduino IDE (which uses an easier version of C++ to write code).

Apparatus:

1) Arduino IDE (any version)
2) Arduino Uno (R3) board or Arduino mega 2560 board
3) LED lights (RED, GREEN, and YELLOW)
4) Three 100 ohms' resistors, and
5) Jumper wires

Circuit Diagram:



(a) (b)

Figure 1: Circuit connections for (a) LED blink test and (b) Traffic light control system

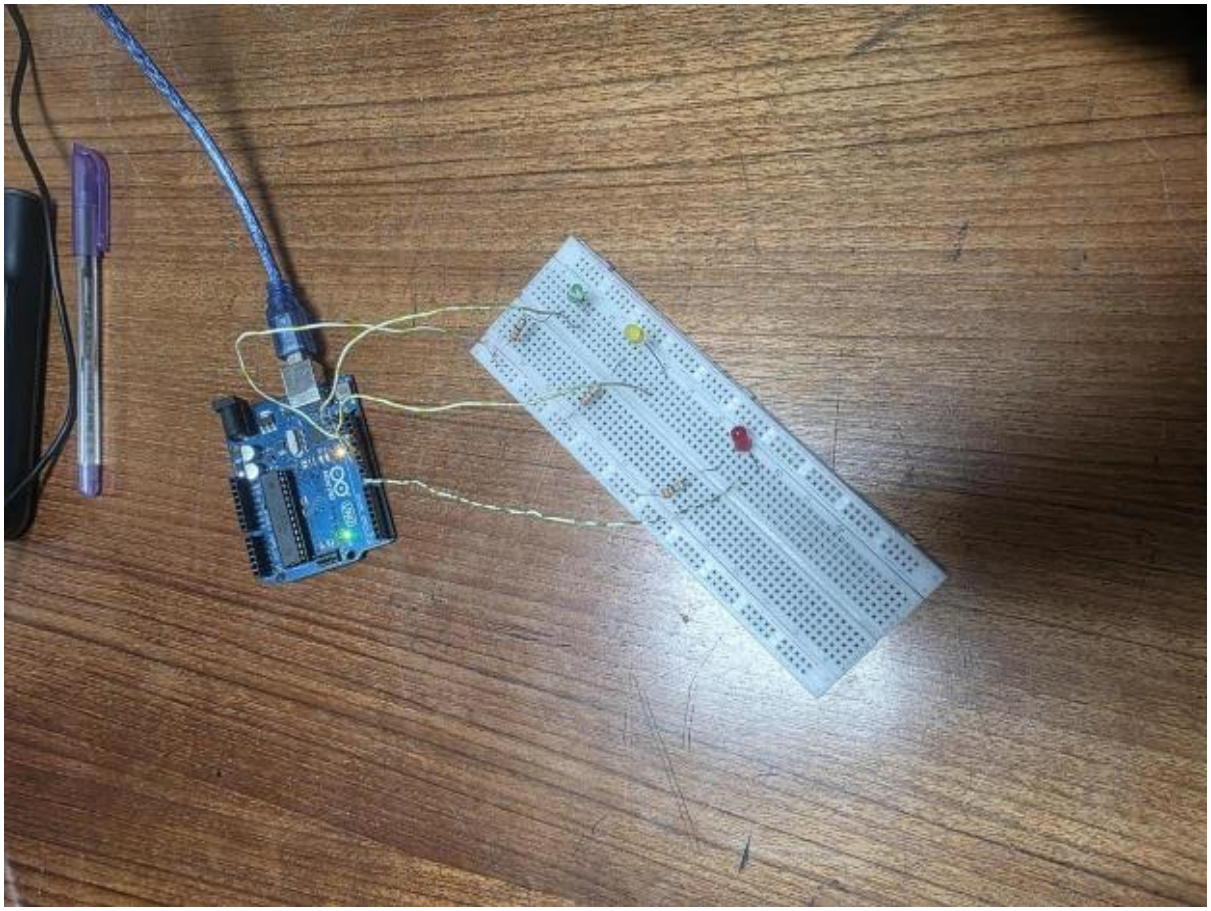
Code / Program:

v

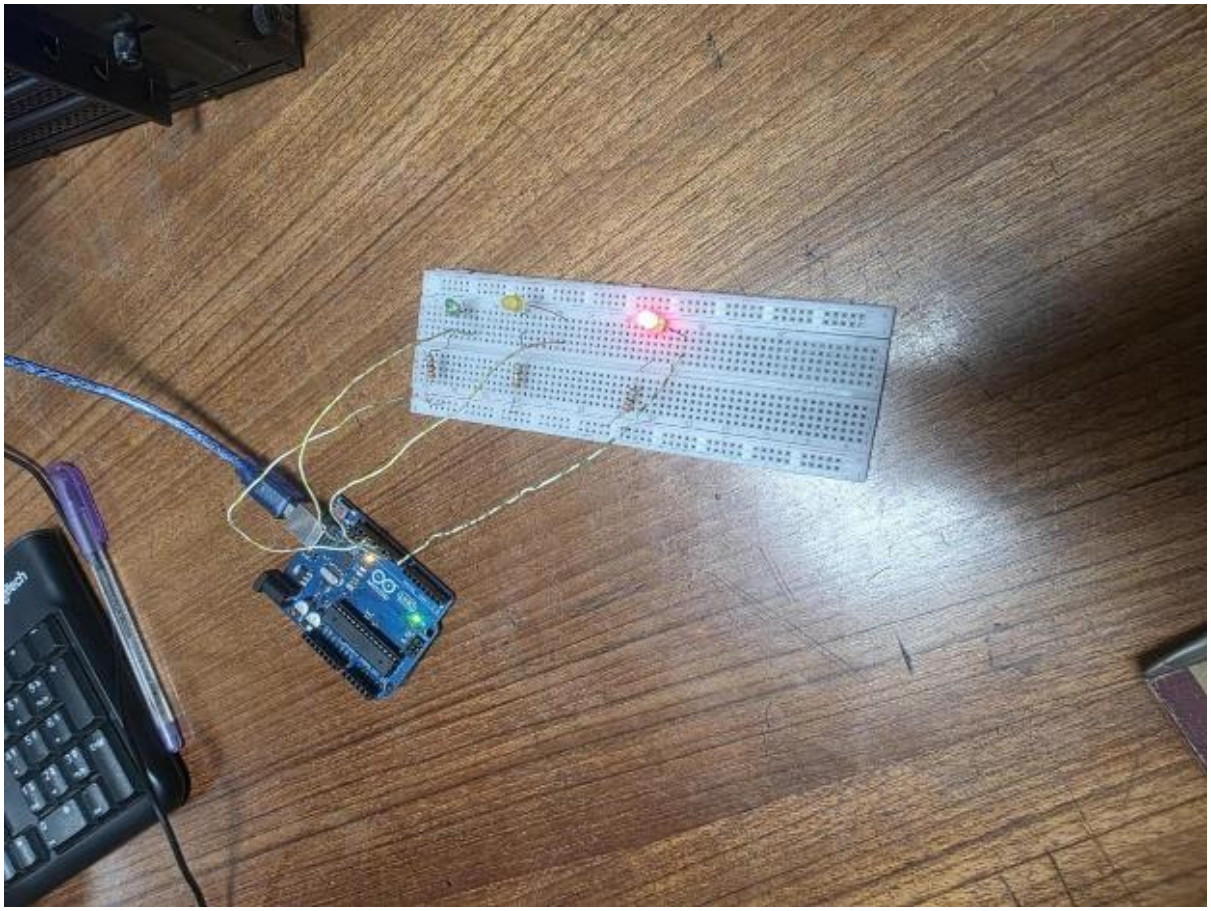
```
void setup() {
  // pin connections for the LED lights
  pinMode(8, OUTPUT);   // Red
  pinMode(10, OUTPUT);  // Yellow
  pinMode(12, OUTPUT);  // Green
}

void loop() {

  // RED ON for 5 seconds
  digitalWrite(8, HIGH);
  delay(5000);
  digitalWrite(8, LOW);
```

```
  // YELLOW ON for 2 seconds
  digitalWrite(10, HIGH);
  delay(2000);
  digitalWrite(10, LOW);

  // YELLOW BLINKS for 3 seconds (3 times)
  for (int i = 0; i < 4; i++) {
    digitalWrite(10, HIGH);
    delay(500);
    digitalWrite(10, LOW);
    delay(500);
    digitalWrite(10, LOW);
  }

  // GREEN stays ON for 5 seconds
  digitalWrite(12, HIGH);
  delay(5000);
  digitalWrite(12, LOW);
}
```

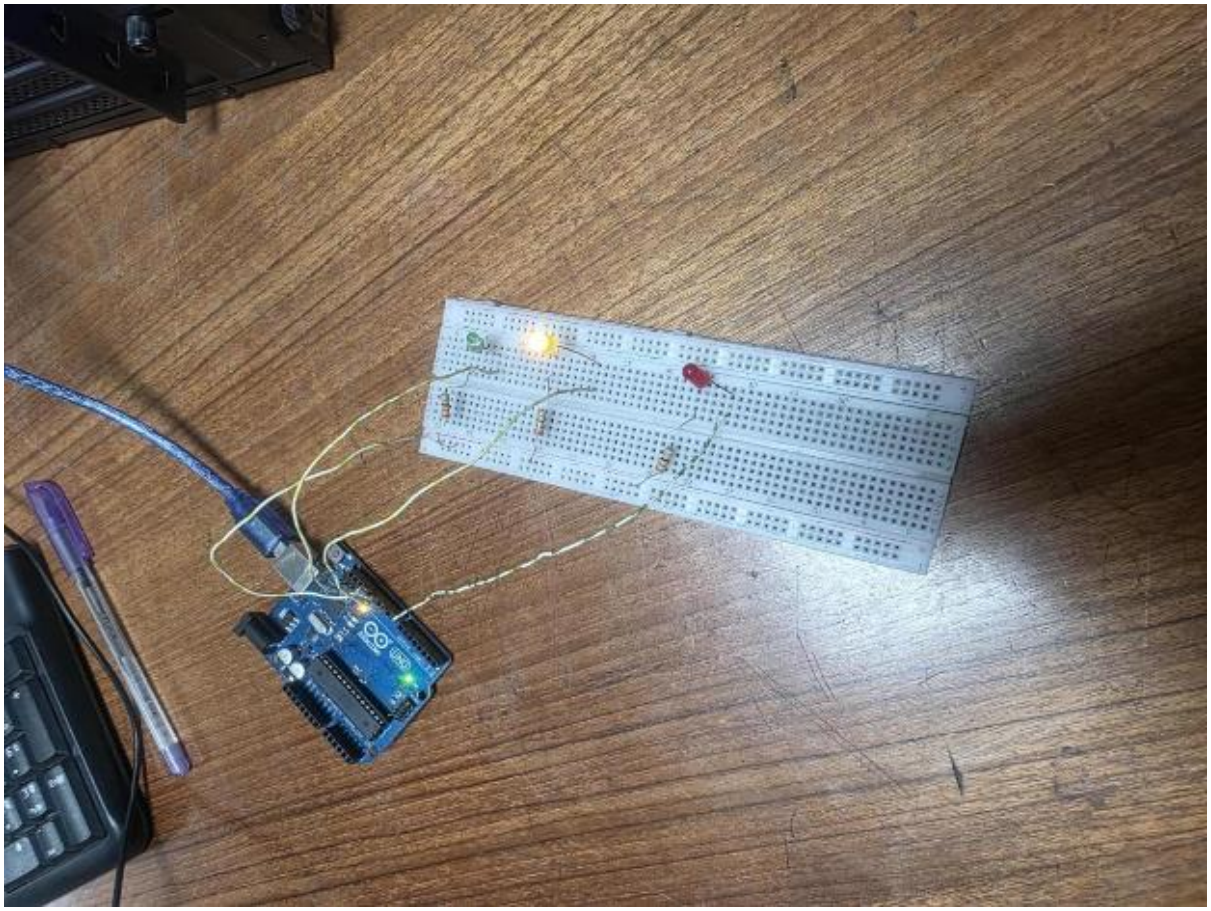
Experimental Output Results:



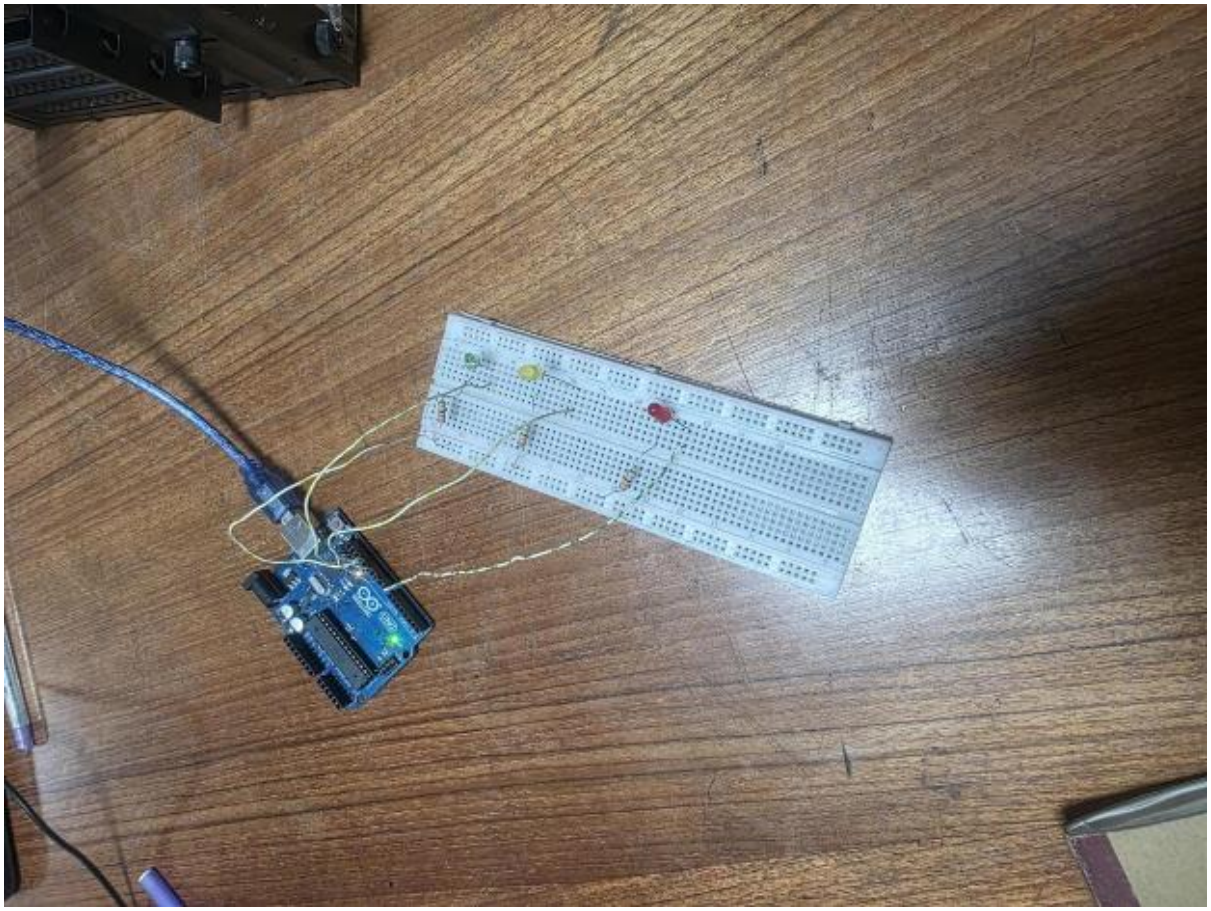
(a)



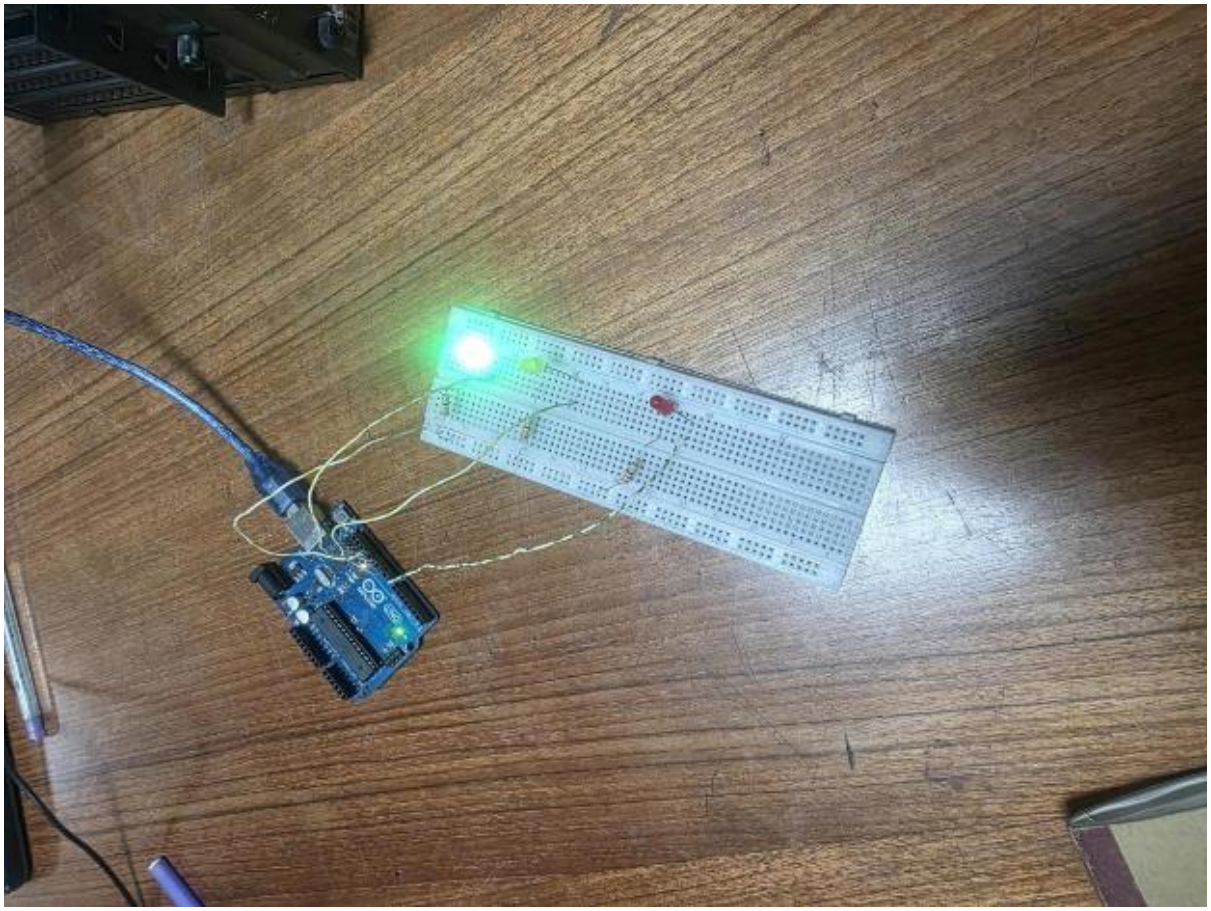
(b)



(c)



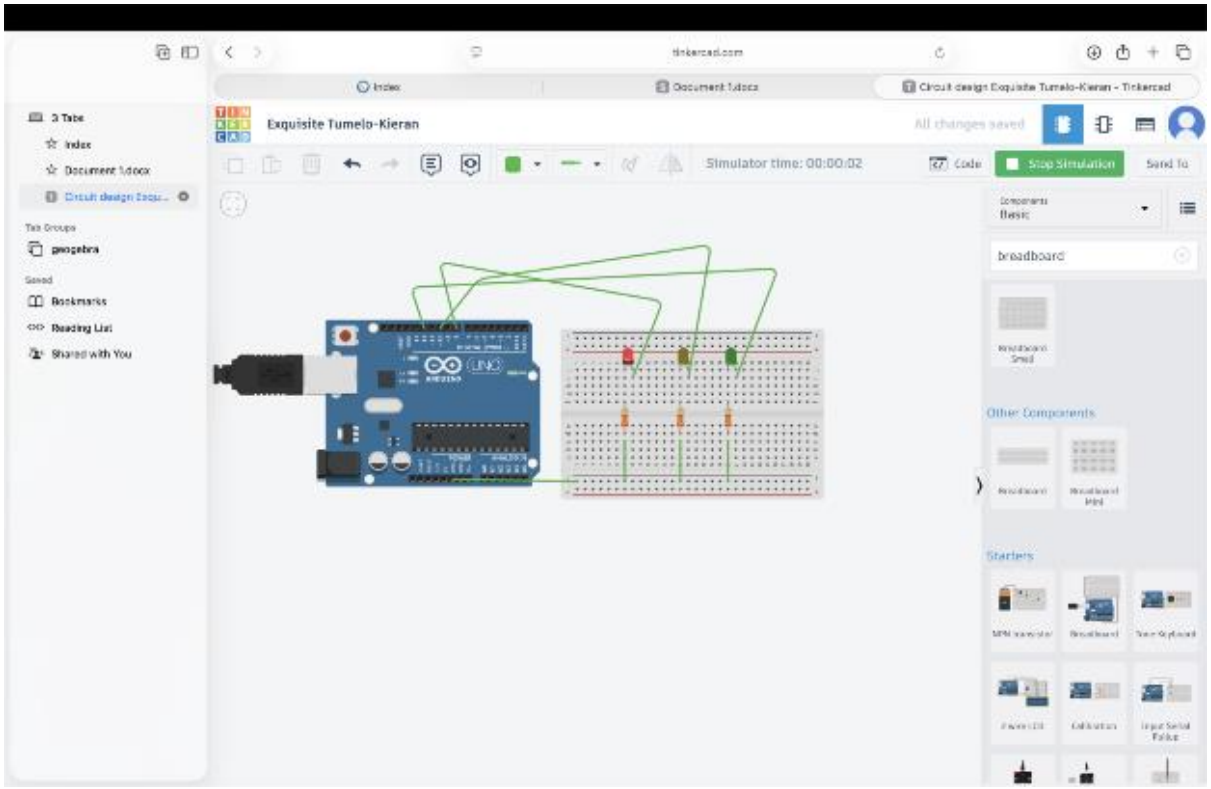
(d)



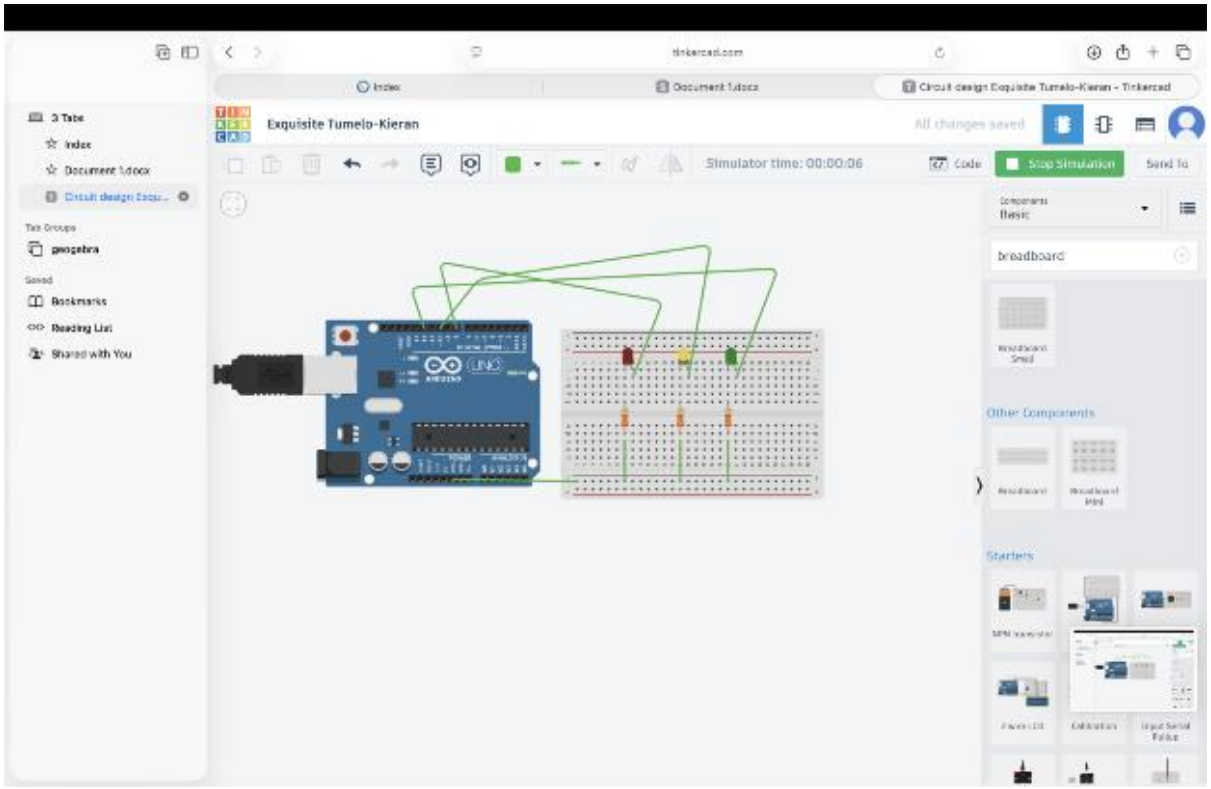
(e)

Figure 2: Circuit connections for Traffic light control system (a) LEDs are LOW (b) red LED is HIGH (c) Yellow LED is HIGH (d) LEDs are LOW (e) green LED is HIGH

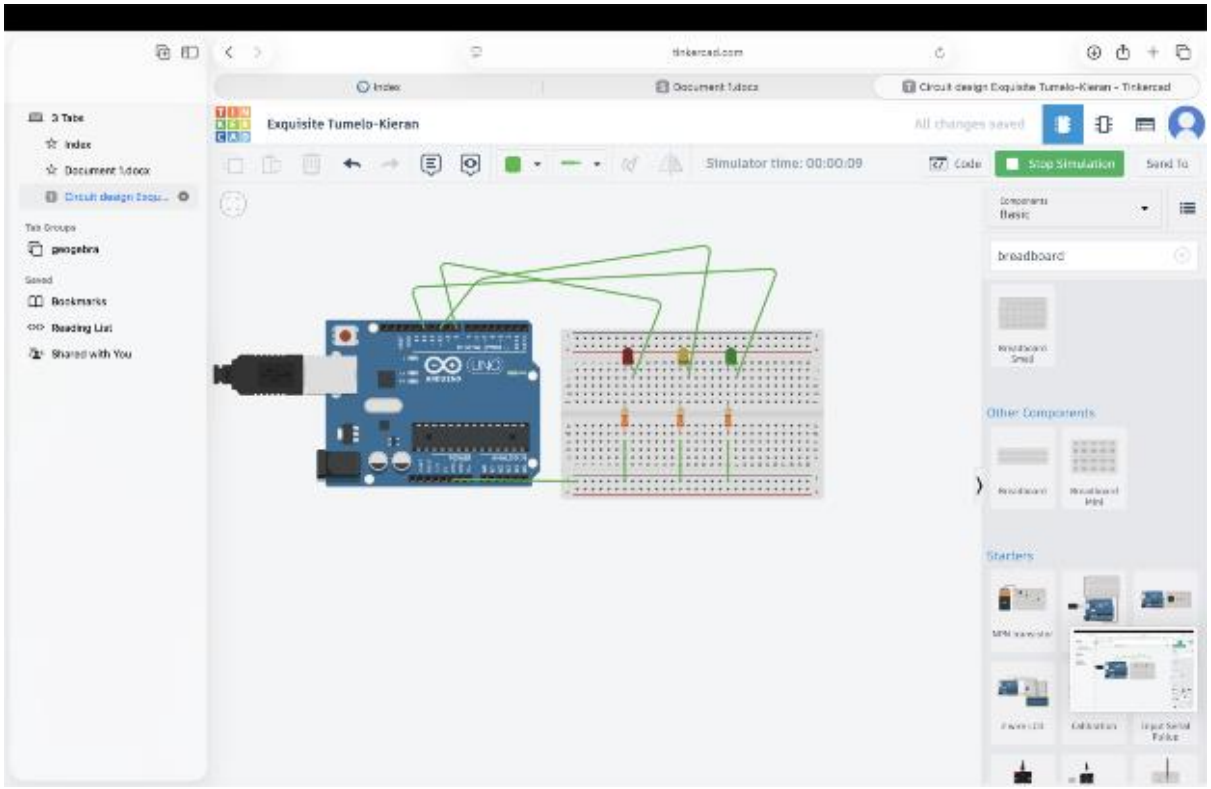
Simulation Output Results:


tinkercad.com
Exquisite Tumelo-Kieran
All changes saved
Simulator time: 00:00:02
Code
Stop Simulation
Send to
Components
Basic
breadboard
Other Components
Starters

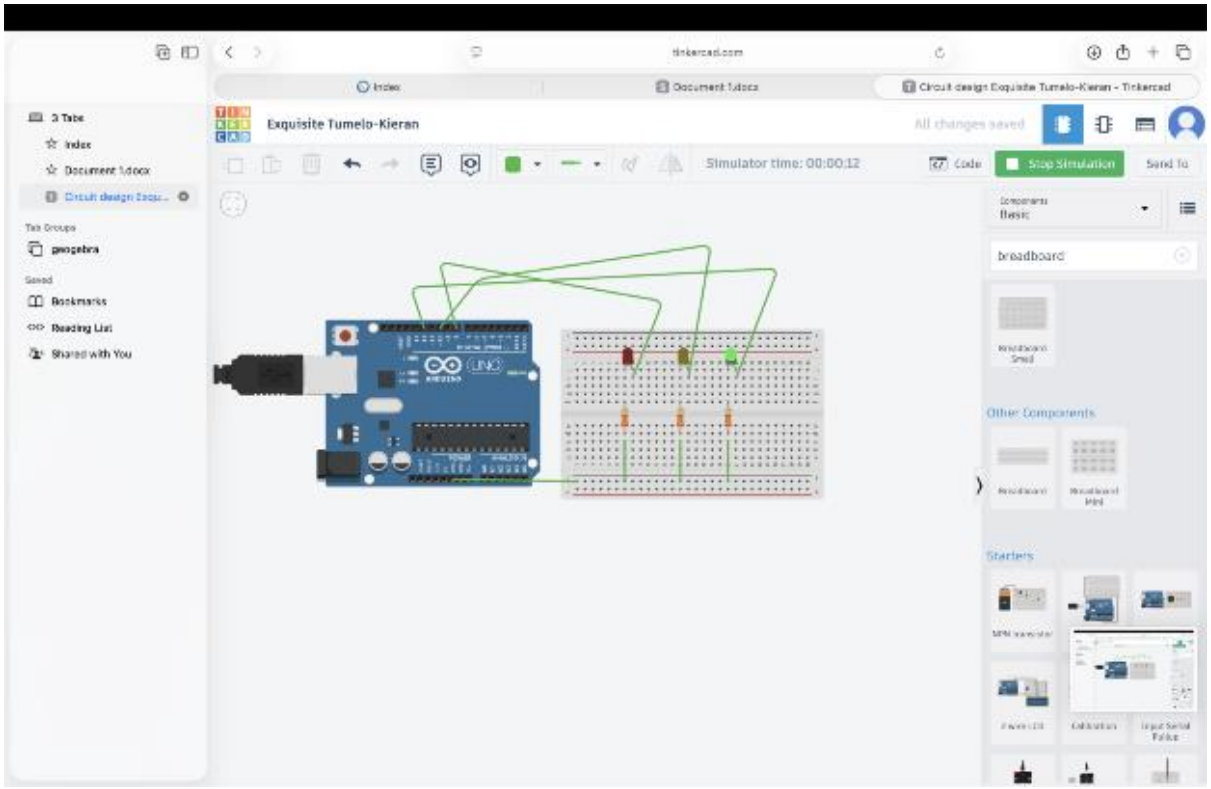
(a)


tinkercad.com
Exquisite Tumelo-Kieran
All changes saved
Simulator time: 00:00:06
Code
Stop Simulation
Send to
Components
Basic
breadboard
Other Components
Starters

(b)


tinkercad.com
Exquisite Tumelo-Kieran
All changes saved
Simulator time: 00:00:09
Code
Stop Simulation
Send to
Components
Basic
breadboard
Other Components
Starters

(c)


tinkercad.com
Exquisite Tumelo-Kieran
All changes saved
Simulator time: 00:00:12
Code
Stop Simulation
Send to
Components
Basic
breadboard
Other Components
Starters

(d)

Figure 3: Circuit connections for Traffic light control system (a) red

LED is HIGH (b) Yellow LED is HIGH (d) LEDs are LOW (e) green LED is HIGH

Answers to the question in the Lab Manual:

1. All the code and the scripts from the Arduino codes of the experiment.

Provided as Code/Program.

2. Output result images:

Provided as Experimental Output Results.

Discussion:

From this experiment we were able to learn about basic principles of a Microcontroller and its working principles using Arduino board by designing and implementing a blinking LED circuit and a traffic light control system. For the blinking test a yellow LED was placed on the breadboard with a 220Ω resistor connected to the positive terminal and ground connected to the negative terminal. The other pin of the resistor was connected to the Arduino board's output pin as per the instructions of the written code. For the traffic light control system three LEDs (red, yellow, and green) were connected to the Arduino board's output pins that are assigned in the written code. Arduino was connected via USB with the PC to power the board and for uploading the code. After carefully implementing the system, the code written on the Arduino IDE was uploaded to the Arduino UNO R3 board via the USB and the system performed as intended completing the objective of this experiment. The same system was implemented on TinkerCad circuit simulation website to perform the simulation. Which gave the same output as the hardware implemented system.

Conclusion:

In this experiment, a traffic light system was successfully simulated using Arduino and LEDs. The program controlled the Red, Yellow, and Green LEDs to follow a realistic traffic sequence, including a blinking Yellowlight before turning solid. The experiment demonstrated the effective use of digital output pins, timing with delay(), and loop control structures in Arduino programming. Through this lab, the principles of sequential control and basic embedded system programming were reinforced, and the project highlighted the importance of clear coding structure for readability and future modifications. Overall, the experiment achieved its goal of modeling a real-world traffic light system in a simple, functional setup.

References:

1) https://www.arduino.cc/.

2) https://www.coursera.org/learn/arduino/lecture/ei4ni/1-10-first-glance-at-a-program.

3) Jeremy Blum; Exploring Arduino: Tools and Techniques for Engineering Wizardry